

FYS4480/9480 Sept 3



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$$\langle p | q \rangle = \langle c | a_p a_q^\dagger | c \rangle$$

$$= \underbrace{\langle c | \delta_{pq} | c \rangle}_{a_p a_q^\dagger} - \underbrace{\langle c | a_q^\dagger a_p | c \rangle}_{N[a_p a_q^\dagger]}$$

$$\langle c | a_p a_q a_s^\dagger | c \rangle$$

$$\begin{aligned} \langle pq | rs \rangle &= \langle c | a_q a_p a_r^\dagger a_s^\dagger | c \rangle \\ &= N[a_q a_p a_r^\dagger a_s^\dagger] + \end{aligned}$$

$$\sum N \left[a_f a_p \overbrace{a_j^+ a_n^+} \right]$$

(1) .

all contractions
of two operators

$$\left(\delta_{fs} N \left[a_p a_n^+ \right] \right) = 0$$

$$+ \sum_{(2)} N \left[a_f a_p \overbrace{a_j^+ a_n^+} \right]$$

$$\overbrace{a_f a_p a_j^+ a_n^+} = \delta_{ps} \delta_{fn}$$

$$\underbrace{a_f a_p a_j^+ a_n^+} = - \delta_{fs} \delta_{pn}$$

Wick's theorem: chain of operators

$$xy z \dots w =$$

$$N[xy z \dots w] + \sum_{(1)} N[\overbrace{xy z \dots}^{\quad} w]$$

$$+ \sum_{(2)} N[\overbrace{xy z \dots}^{\quad} \overbrace{\quad}^{\quad} w]$$

$$+ \dots \sum_{(\frac{M}{2})} N[\overbrace{xy z \dots}^{\quad} \overbrace{\quad}^{\quad} \overbrace{\quad}^{\quad} \dots w]$$

contractions

$M = \# \text{ operators}$

$$\langle 0 | xy z \dots w | 0 \rangle = \sum_{(M/2)} N[\overbrace{xy z \dots}^{\quad} \overbrace{\quad}^{\quad} \overbrace{\quad}^{\quad} \dots w]$$

$\hat{N}$ = number operator

$$\hat{N} = \sum_p a_p^\dagger a_p$$

$$\hat{N} |12, 3, \dots, N\rangle = N |12, 3, \dots, N\rangle$$

$$N=2 \quad a_1^\dagger a_2^\dagger |0\rangle$$

$$\langle 12 | \hat{N} | 12 \rangle =$$

$$\sum_p \langle 0 | a_2 a_1 a_p^\dagger a_p a_1^\dagger a_2^\dagger | 0 \rangle$$

$$= \sum_3 \underbrace{N [a_2 a_1 a_p^\dagger a_p a_1^\dagger a_2^\dagger]}_{\text{contractions}}$$

$$\underbrace{a_2 a_1 a_p^+ a_p a_1^+ a_2^+}_{\text{}} = \delta_{p1} \delta_{22}$$

$$\underbrace{a_2 a_1 a_p^+ a_p}_{\text{}} \underbrace{a_1^+ a_2^+}_{\text{}} = \delta_{p2} \delta_{11}$$

$$\sum_{p, i} \delta_{p, i} \quad \text{here } N=2 \quad \sum$$

$$= \sum_{p=1}^d \sum_{i=1}^N \delta_{p, i} = N$$

General one-body operator

$$\hat{h}_0(x) \rightarrow \sum_{pq} \langle p | \hat{h}_0 | q \rangle a_p^\dagger a_q$$

in general

$$\hat{O} = \sum_{pq} \langle p | \hat{O} | q \rangle a_p^\dagger a_q$$

$$\langle p | \hat{O} | q \rangle = \int dx \psi_p^*(x) \hat{O} \psi_q(x)$$

$$\hat{h}_0 | q \rangle = \epsilon_q | q \rangle$$

$$\langle p | \hat{h}_0 | q \rangle = \epsilon_q \delta_{pq}$$

$$\hat{H}_0 = \sum_p \left(-\frac{\nabla_p^2}{2m} + u_{\text{ext}}(x_p) \right)$$

$$\hat{H}_0 = \sum_p \varepsilon_p a_p^\dagger a_p$$

$$\langle 12 | \hat{H}_0 | 12 \rangle$$

$$= \sum_p \varepsilon_p \langle 0 | a_2 a_1 a_p^\dagger a_p a_1^\dagger a_2^\dagger | 0 \rangle$$

$$= \varepsilon_1 + \varepsilon_2$$

$$\sum_{i' < j'} v(x_{i'}, x_{j'}) = \sum_{i' < j'} v(|x_{i'} - x_{j'}|)$$

$$\rightarrow \frac{1}{2} \sum_{pqrs} \langle \underline{pq} | \underline{v} | \underline{rs} \rangle a_p^\dagger a_q^\dagger a_s a_r$$

$$\langle \underline{pq} | \underline{v} | \underline{rs} \rangle = \int dx dy \frac{\phi_p^*(x) \phi_q^*(y)}{v(x, y) \phi_p(x) \phi_q(y)}$$

$$\sim 100 \text{ } \underline{pq} \text{ configs} \rightarrow \sqrt{IJ}$$

$$\frac{1}{4} \sum_{pqrs} \underbrace{\langle pq | v | rs \rangle}_{AS} a_p^\dagger a_q^\dagger a_s a_r$$

$$\langle pq | v | rs \rangle - \langle pq | v | sr \rangle$$

$$\underbrace{\langle 12 |}_{AS} \underbrace{H_I}_{SD} | \underbrace{12 \rangle}_{AS} = \underbrace{\langle 12 |}_{SD} \underbrace{v | 12 \rangle}_{AS} \text{ Direct term}$$

$$- \langle 12 | v | 21 \rangle$$

Exchanging
term

2nd - Quantum field theory

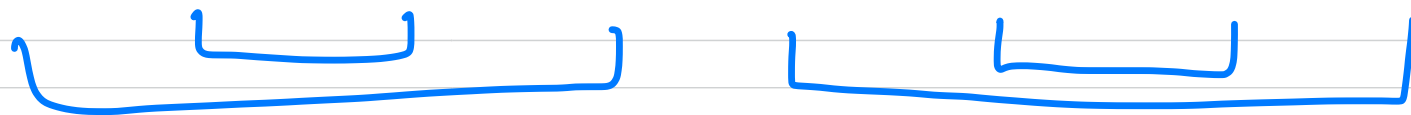
$$\langle 12 | H | 12 \rangle =$$

$$\frac{1}{4} \sum_{pqrs} \langle pq | v | rs \rangle_{AS}$$

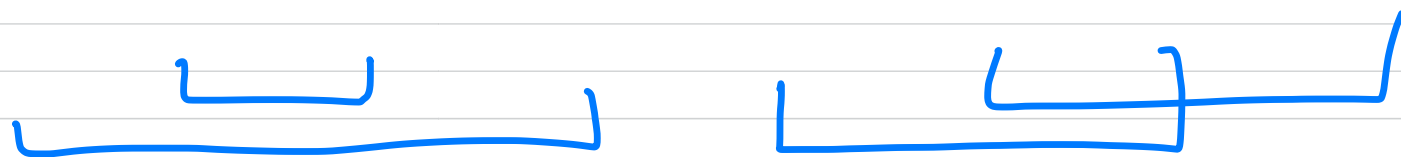
$$\times \langle 0 | a_2 a_1 a_p^\dagger a_q^\dagger a_r a_s a_1^\dagger a_2^\dagger | 0 \rangle$$

δ_{2q}
 δ_{1p}
 δ_{s2}
 δ_{r1}

$$a_2 a_1 a_p^+ a_q^+ a_5 a_2 a_1^+ a_2^+$$



$$\langle 12 | \alpha | 12 \rangle_{A5} \cdot \frac{1}{4}$$



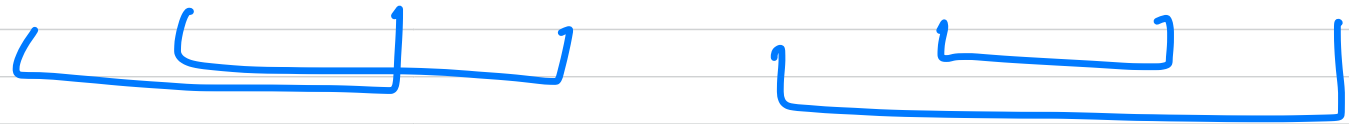
$$\delta_{2q} \delta_{1p}$$

$$- \delta_{s1} \delta_{r2}$$

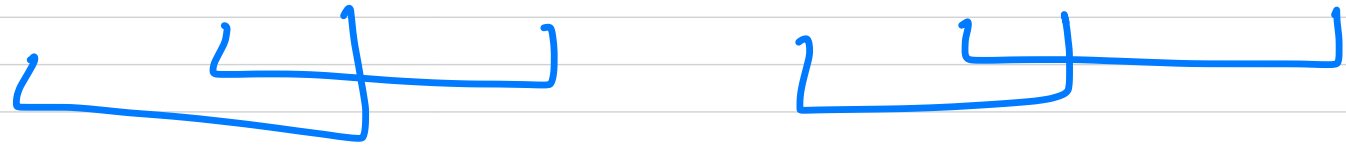
$$- \langle 12 | \alpha | 21 \rangle_{A5} \frac{1}{4}$$

$$= + \langle 12 | \alpha | 22 \rangle_{A5} \frac{1}{4}$$

$$a_2 a_1 a_p^+ a_q^+ a_5 a_2 a_1^+ a_2^+$$



$$- \langle 211\bar{1}12 \rangle_{A_5} \frac{1}{4} = \langle 221\bar{1}12 \rangle_{A_5} \frac{1}{4}$$



$$\langle 211\bar{1}12 \rangle_{A_5} \frac{1}{4} =$$

$$\langle 121\bar{1}12 \rangle_{A_5} \frac{1}{4}$$

$$\Rightarrow \langle 121\bar{1}12 \rangle_{A_5}$$

$$= \langle 121\bar{1}12 \rangle - \langle 121\bar{1}21 \rangle$$